Higher-Order Structure in Residue-Class Transitions of Consecutive Primes

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https://github.com/thinkthoughts/prime-numbers-lab

Abstract

We encode consecutive primes as residue classes modulo 30, producing a sequence over 8 states $\{1, 7, 11, 13, 17, 19, 23, 29\}$. We construct a first-order transition operator P and compare the two-step operator $P^{(2)}$ to the baseline P^2 via $\Delta = P^{(2)} - P^2$.

Using the first N primes, Δ has low-rank structure. Low-rank approximations improve prediction of $P^{(2)}$ relative to P^2.

Synthetic controls produce distinct residual operators. Residue-class transitions of consecutive primes modulo 30 are represented by the residual operator $\Delta = P^{(2)} - P^2$, which exhibits higher-order structure.

1 Introduction

We study transitions of consecutive primes. Reducing them modulo 30 yields a sequence over 8 residue states $\{1, 7, 11, 13, 17, 19, 23, 29\}$.

We construct a transition operator P and compare $P^{(2)}$ to P^2. The residual

$$\Delta = P^{(2)} - P^2$$

isolates second-order structure.

We test whether Δ reflects random variation or systematic structure.

2 Related Work

Classical models of prime gaps include probabilistic heuristics such as Cramér's model [3] and analytic frameworks for prime distributions [4].

Markov models provide a baseline for transition structure [1], while information-theoretic measures quantify higher-order structure relative to first-order statistics [2].

This work analyzes residue-class transitions of consecutive primes using empirical transition operators and the residual operator $\Delta = P^{(2)} - P^2$.

3 Transition Framework

Let $\{x_n\}_{n=1}^N$ be residues modulo 30 of the first N primes.

We estimate

$$P_{ij} = \frac{\#\{n : x_n = i, x_{n+1} = j\}}{\#\{n : x_n = i\}}.$$

Define

$$\Delta = P^{(2)} - P^2.$$

4 Results

Figure 1 shows the transition operator P.

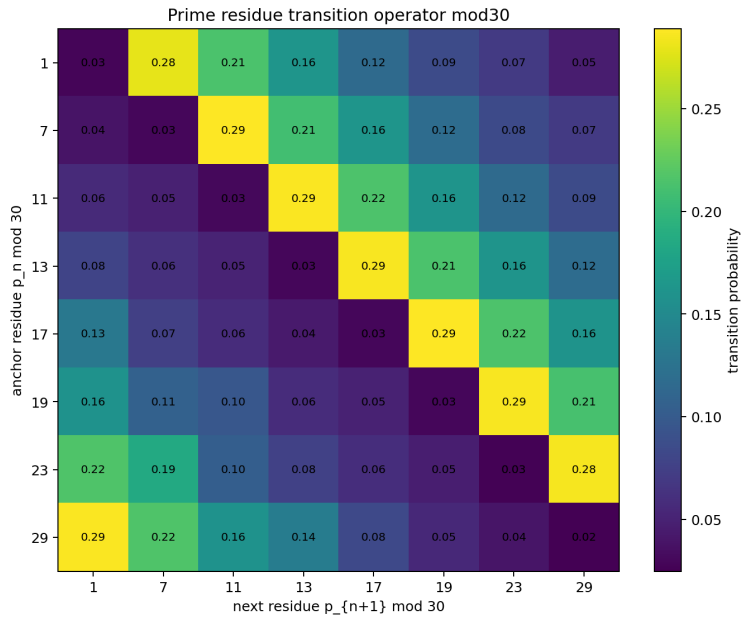

Figure 1: First-order transition operator P.

Figure 2 shows the residual $\Delta = P^{(2)} - P^2$, with nonuniform deviations.

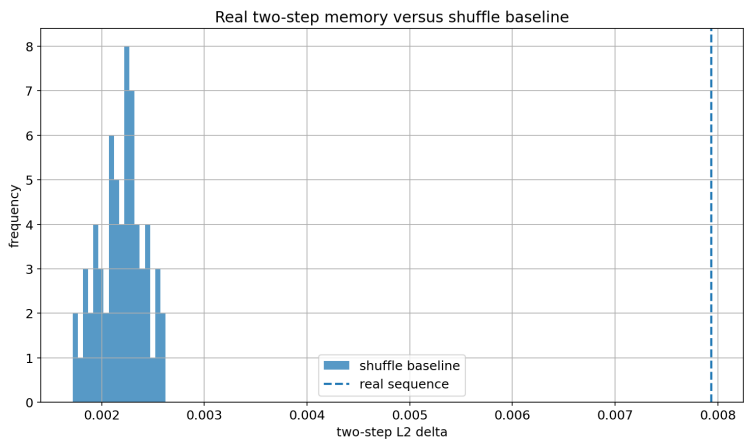

Figure 2: Residual operator $\Delta = P^{(2)} - P^2$.

Figure 3 shows low-rank structure in Δ.
Figure 4 shows statistical validation across controls.
Figure 5 shows summary ratios for empirical and control statistics.

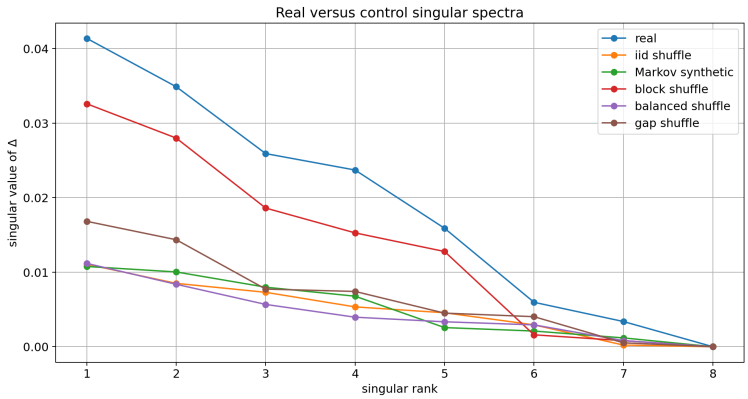

Figure 3: Low-rank structure of Δ.

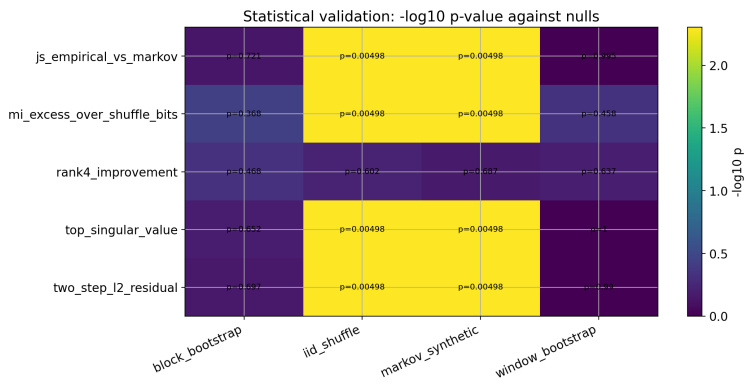

Figure 4: Statistical validation p-values.

5 Validation

The prime sequence differs from iid and Markov control sequences in the Δ. Block resampling shows partial agreement, indicating local structure.

6 Discussion

The operator Δ isolates second-order structure relative to P. Its is low-rank, with deviations concentrated in a small number of components.

A small number of components approximates $P^{(2)}$ more accurately than P^2.

This structure persists across sample sizes N.

7 Conclusion

Residue-class transitions of consecutive primes modulo 30 are represented by the residual operator $\Delta = P^{(2)} - P^2$, which exhibits higher-order structure.

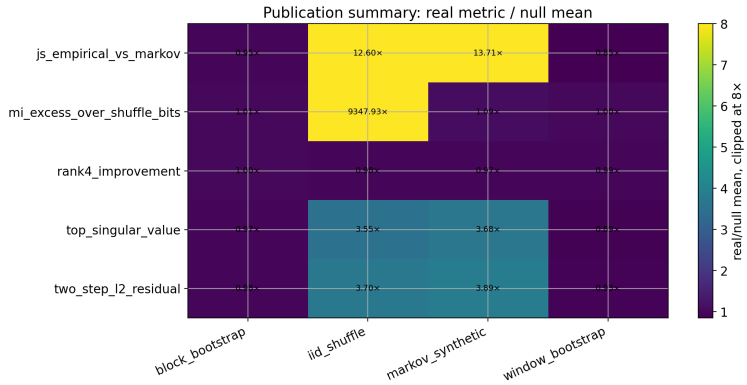

Figure 5: Summary ratios for empirical and control statistics.

References

[1] David Aldous. Random walks on finite groups and rapidly mixing markov chains. *Seminaire de Probabilites*, 1983.

[2] Thomas Cover and Joy Thomas. *Elements of Information Theory*. Wiley, 2006.

[3] Harald Cramér. On the order of magnitude of prime gaps. *Acta Arithmetica*, 1936.

[4] G. H. Hardy and E. M. Wright. *An Introduction to the Theory of Numbers*. Oxford University Press, 1979.